

AI Tools Used

Date	15 JUNE 2026
Team ID	XXXXXX
Project Name	Sustainable Waste Management Assistant Using Generative AI
Maximum Marks	2 Marks

AI Tools Used:

Documenting the specific artificial intelligence tools and platforms utilized during your project lifecycle helps evaluate efficiency and technological integration. Detail each AI tool, how it was applied, and the tangible benefit it provided to your workflow or final product.

S · N o	AI Tool / Platform Name	Specific Use Case / Application	Project Phase Applied	Impact / Benefit
1	Groq API (LLaMA-3.3-70B Versatile)	Classified waste items, generated disposal instructions, hazard warnings, and eco-friendly suggestions	Development Phase	Enabled accurate, real-time waste classification and guidance
2	hart.js (AI-assisted integration)	Created interactive dashboards for waste analytics (recyclable vs non-recyclable, scan frequency, hazardous items)	Visualization Phase	Improved user engagement with clear sustainability insights

3	Leaflet.js + OpenStreetMa	Rendered maps of nearby collection centers with color-coded markers	Deployment Phase	provided free, reliable geolocation services without paid APIs
4	Copilot / Generative AI Assistant	Assisted in prompt engineering, boilerplate code generation, and refining system requirements	Design & Development Phase	Accelerated development, reduced coding effort, and improved documentation quality
5	Firebase Firestore (AI-ready cloud DB)	Stored scan history, user activity records, and analytics data	Data Management Phase	Delivered scalable, serverless storage with real-time sync